Analytics Capstone Project (ANCP)

Sparta: Wardrobe Demand Analytics for Brands

Deliverable D1: Project Charter

Team number	7
Deliverable code / version	D1 / v1
Date	13 October 2026 (due Fri 16 October 2026, 17:00 IST)
Lead owner for this deliverable	Ambarish Sengupta
Term	Term 5, 6 October to 20 November 2026
Members	C. Prashanth (BA05015) Arpan Shah (BA05009) Ambarish Sengupta (BA05007)

1. Problem statement

Business context. Sparta is a free digital wardrobe for men. Users catalogue the clothes they own, plan outfits, record purchases and keep a wishlist. With each user's opt-in consent, anonymised and aggregated data is shared with brands and retailers. Today these vendors plan ranges and stock largely on past sales and judgement, which cannot show what customers already own or still want.

Business decision (primary). Which menswear categories, colours and fits should a brand add or expand in its next-season range?

Decision-maker. The head of product and design at a menswear brand. The secondary decision, how much stock to commit by category, belongs to a retailer's inventory and merchandising manager.

Analytics question (one measurable sentence). Can a demand-gap model built from aggregated ownership, purchase and wishlist signals, evaluated on a time-based holdout, rank next-season menswear category-colour-fit segments with at least 15% lower weighted absolute percentage error (WAPE) than a same-period-last-year baseline, while keeping the top-10 recommended segments at 60% precision or higher?

2. Stakeholder map

Role	Who	Interest / involvement
Decides	Head of product and design (menswear brand)	Uses the ranked gaps to set the next-season range. Primary decision-maker.
Decides (secondary)	Inventory and merchandising manager (retailer)	Uses category-level demand forecasts to plan stocking.
Affected	Sparta users (men)	Better-fitting ranges and products; data used only with opt-in consent, only in aggregate.
Affected	Brands and retailers (vendors)	Pay for dashboards and promotions; carry the cost of wrong forecasts.
Supplies data	Public fashion datasets (Section 5)	Transactions, article attributes and outfit data; terms of use apply.
Supplies data	Sparta team (simulated app data)	Synthetic Sparta-style wardrobe, wishlist and purchase records.
Guides	ANCP faculty	Feedback on each milestone; marks the deliverables.

Table 1. Stakeholder map.

3. Success metrics and acceptance criteria

Metric	Baseline compared against	Target	Acceptance test
WAPE of 8-week demand forecast per category-colour-fit segment	Same-period-last-year (naive seasonal) forecast	15% or more lower than baseline	Computed on a time-based holdout never used in training or tuning
Precision@10 of recommended gap segments	Top-10 segments by previous-season sales	0.60 or higher, and above the baseline	A recommended segment counts as correct if its realised demand growth lands in the holdout top quartile
Prediction-interval coverage	Nominal 80% interval	Between 75% and 85%	Share of holdout actuals inside the interval
Reproducibility	Fresh clone of the repository	Main result reruns end to end	One command regenerates the tables and figures, with versioned data
Decision usability	A one-page brief without the model details	A non-analyst names the decision and the top three gaps	Read-through by one reader outside the team

Table 2. Success metrics, baselines, targets and acceptance tests.

The project succeeds when the first three metrics meet their targets together on the holdout and the last two pass. If the forecast does not beat the baseline, the final report will say so and state by how much.

4. Scope

In scope.

- Menswear categories (tops, bottoms, outerwear, footwear, accessories) by colour, fit and season.
- Aggregated demand forecasting and wardrobe-gap analysis at segment level.
- Time-based validation, a baseline, uncertainty estimates and a one-page decision brief.
- A vendor-facing results view in the final report (a full dashboard is not built).

Out of scope.

- Building the Sparta mobile or web app, or its backend.
- Womenswear and other audiences; outfit recommendation or inspiration-matching models.
- Individual-level personalisation and any use of identifiable data.
- Real-time serving; a production deployment (D6 plans one on paper only).

Assumptions.

- Public retail transactions are a usable stand-in for purchase demand, and wishlist behaviour can be simulated from them with documented rules.
- Opt-in consent leaves enough users in each reported segment; segments under 50 users are suppressed.
- Python (pandas, scikit-learn, statsmodels) is sufficient; no paid data is needed.
- Each member can give about 10 hours a week through 20 November 2026.

5. Data plan

Candidate source	Content	Access and rights	Expected size
H&M Personalized Fashion Recommendations (Kaggle, 2022)	Transactions 2018 to 2020, article attributes (type, colour, group), hashed customer IDs	Free download with a Kaggle account; competition rules limit use to non-commercial research and education, which fits this course; terms recorded in the D2 datasheet	About 31.8 million transactions, 105 thousand articles, 1.37 million customers; menswear is a subset
Polyvore Outfits (Vasileva et al., 2018)	Outfit compositions and item images	Public research release; licence checked in D2	Tens of thousands of outfits
Google Trends (menswear terms)	Weekly search interest by region	Public, free, aggregated	A few hundred weekly series
Simulated Sparta data (team-generated)	Wardrobe, wishlist and purchase records following the Sparta schema and fixed taxonomy	Owned by the team; generator code and seeds kept in the repository	About 5,000 simulated users and 300 thousand items

Table 3. Candidate data sources.

Privacy and consent. The repository will hold no personal or identifiable data. Customer IDs in public data are already hashed and will not be linked to other sources. The team will not attempt re-identification, and will record consent and purpose for each source as the DPDP Act, 2023 requires. The simulated data mirrors Sparta's production rules: opt-in consent, aggregation only, and suppression of segments with fewer than 50 users.

6. Feasibility and risks

The project is feasible with public data and Python tools in six weeks. The five largest risks follow.

#	Risk	Likelihood / impact	Mitigation
1	Sparta has no real users yet, so results rest on proxy and simulated data.	High / High	Say so in every deliverable; test on public data; keep the simulation rules simple, documented and seeded; report sensitivity to them.
2	Purchases are not wardrobe gaps, so the target construct is weakly observed.	Medium / High	Define the gap metric in D2; check it on the holdout against realised demand growth; report precision@10 against the previous-season baseline.
3	Time leakage or overfitting inflates the forecast results.	Medium / High	Split by time only; fit all preprocessing inside the training window; keep the final holdout untouched until D5.
4	Licence or privacy terms of a source restrict use.	Low / Medium	Read the terms before download; record them in the datasheet; fall back to the simulated and Trends data.
5	Three people, tight dates and overlapping milestones.	Medium / Medium	Keep scope fixed; use the 7 to 9 November break for revision; one lead per deliverable; review the plan in each weekly meeting.

Table 4. Top five risks and mitigations.

7. Timeline

Code	Deliverable	Due (17:00 IST unless noted)	Lead owner
D1	Project charter	Fri 16 Oct	Ambarish Sengupta
D2	Data audit report, datasheet and cleaned, versioned dataset	Fri 23 Oct	C. Prashanth
D3	EDA notebook and two-page insight summary	Wed 28 Oct	C. Prashanth
D4	Modelling checkpoint (baseline, candidates, experiment log)	Mon 2 Nov	Ambarish Sengupta
D5	Evaluation memo	Thu 5 Nov	Arpan Shah
D6	Deployment and monitoring plan	Fri 6 Nov	Arpan Shah
D7	Draft report and model card; dry-run presentation	Wed 11 Nov (written); Thu 12 Nov (dry run)	C. Prashanth
E1	Final project report	Tue 17 Nov	Ambarish Sengupta
E2	Final slide deck (PDF and PPTX)	Wed 18 Nov, 12:00	Arpan Shah
E2	Final presentation with individual viva	From 24 Nov	Arpan Shah (all present)

Table 5. Milestones D1 to E2 with dates and lead owners.

8. Team roles

Each member leads three deliverables, which meets the rule that every member leads at least two. The lead owns the content, the quality check and the on-time upload; the other two members review before submission.

Member	Roll no.	Leads	Main strengths used
Ambarish Sengupta	BA05007	D1, D4, E1	Product definition, modelling, report writing
C. Prashanth	BA05015	D2, D3, D7	Data audit and cleaning, exploratory analysis, model card
Arpan Shah	BA05009	D5, D6, E2	Evaluation, deployment planning, presentation

Table 6. Lead owners by member.

Appendix A. Meeting log

Date	Attendees	Decisions	Actions
13 Oct 2026 (Meeting 1: product definition and charter planning)	C. Prashanth, Arpan Shah, Ambarish Sengupta	1. Product is a free digital wardrobe for men with an opt-in, aggregated insights layer for brands and retailers. 2. Launch scope: men only, global, Python backend, mobile and web on one backend. 3. Vendor data is opt-in, aggregated, with minimum group sizes. 4. Primary decision is product design and planning; inventory and stocking is secondary. 5. Project title: Sparta: Wardrobe Demand Analytics for Brands.	1. Write and submit the D1 charter (Ambarish Sengupta). 2. Create the team repository with a README (Ambarish Sengupta). 3. Shortlist and read the terms of the public datasets (C. Prashanth). 4. Draft the simulation rules for Sparta-style data (Arpan Shah).

Table A1. Meeting log (date, attendees, decisions, actions).

Appendix B. Responsibility matrix

L = lead owner, R = reviewer, C = consulted on request.

Deliverable	Ambarish Sengupta	C. Prashanth	Arpan Shah
D1 Project charter	L	R	R
D2 Data audit and dataset	R	L	R
D3 EDA and insight summary	R	L	C
D4 Modelling checkpoint	L	R	R
D5 Evaluation memo	R	C	L
D6 Deployment and monitoring plan	C	R	L
D7 Draft report, model card, dry run	R	L	R
E1 Final report	L	R	R
E2 Slides and viva	R	R	L

Table B1. Responsibility matrix.

Appendix C. AI-tool disclosure

The team used Claude (Anthropic) to structure this charter and draft its text from the team's meeting discussion and the course guide. The team checked all decisions, dates, roles and targets against the ANCP Student Guide and agreed them among the three members.

Appendix D. References

H&M Group. (2022). *H&M personalized fashion recommendations* [Data set]. Kaggle. https://www.kaggle.com/competitions/h-and-m-personalized-fashion-recommendations

Vasileva, M. I., Plummer, B. A., Dusad, K., Rajpal, S., Kumar, R., & Forsyth, D. (2018). Learning type-aware embeddings for fashion compatibility. *Proceedings of the European Conference on Computer Vision (ECCV)*, 390-405.

Digital Personal Data Protection Act, 2023 (India).